

Semi-parametric robust mean estimation based on provable quantile inequality

Tuban Lee^{a,1}

^a Institute of Biomathematics, Macau SAR 999078, China

This manuscript was compiled on January 27, 2023

As arguably the most fundamental problem in statistics, non-parametric robust location estimation has many prominent solutions, such as the trimmed mean, Winsorized mean, Hodges–Lehmann estimator, and median of means. Recent research suggests that their biases can be quite different. However, the underlying principles remain largely unclear. Here, similar to the mean-median-mode inequality, it is proven that the quantile inequality holds for nearly all common unimodal distributions. Further deductions explain why the Winsorized mean and median of means generally have smaller biases compared to the trimmed mean. Building on the ν th U -quantile inequality, binomial Hodges–Lehmann mean is proposed as the bias-optimum semi-parametric robust mean estimator.

semi-parametric | non-parametric | asymptotic | unimodal |
Hodges–Lehmann estimator

In 1823, Gauss (1) proved that for any unimodal distribution with a finite second moment, $|m - \mu| \leq \sqrt{\frac{3}{4}}\omega$, where μ is the population mean, m is the population median, ω is the root mean square deviation from the mode, M . Bernard, Kazzi, and Vanduffel (2020) (2) derived bias bounds for the ϵ -symmetric quantile average (SQA_ϵ) for unimodal distributions, building on the work of Karlin and Novikoff (1963) and Li, Shao, Wang, and Yang (2018) (3, 4). They showed that the m has the smallest maximum distance to the μ among all symmetric quantile averages. Daniell, in 1920, (5) analyzed a class of estimators, weighted averages (WA_ϵ), which are linear functions of order statistics, and identified that ϵ -symmetric trimmed mean (TM_ϵ) belongs to this class. Another popular choice, the ϵ -symmetric Winsorized mean (WM_ϵ), which was named after Winsor and introduced by Tukey (6) and Dixon (7) in 1960, is also a type of weighted average according to Daniell's definition (5). Without assuming unimodality, Bieniek (2016) derived exact bias upper bounds of the Winsorized mean based on Danielak and Rychlik's work (2003) on the trimmed mean and confirmed that the former is smaller than the latter (8, 9). In 1963, Hodges and Lehmann (10) proposed a class of nonparametric location estimators based on rank tests and deduced the median of pairwise means as a robust location estimator for a symmetric population based on the Wilcoxon signed-rank statistic (11). The concept of median of means (MoM) was implicit several times in Nemirovsky and Yudin (1983) (12), Jerrum, Valiant, and Vazirani (1986), (13) and Alon, Matias and Szegedy (1996) (14)'s works. Devroye, Lerasle, Lugosi, and Oliveira (2016) showed that MoM nears the optimum of mean estimation with regards to concentration bounds when the distribution has a heavy tail (15). Here, the ϵ -stratified mean is defined as

$$\text{SM}_{\epsilon,n} := \frac{3}{n} \left(\sum_{j=1}^{\frac{1}{3\epsilon}} \sum_{i_j=(3j-2)n\epsilon+1}^{(3j-1)n\epsilon} X_{i_j} \right),$$

where $X_1 \leq \dots \leq X_n$ denote the order statistics of a sample of n independent and identically distributed random variables $X_1, \dots, X_n$, $\frac{1}{\epsilon} \bmod 3 = 0$, $\frac{1}{\epsilon} \geq 9$. If the subscript n is omitted, only the asymptotic behavior is considered. The basic idea is to divide the random variables into three blocks according to their order, and then compute the mean of the middle block, which is the median of all three blocks. Thus, it is essentially a deterministic version of MoM. The exact solution for $n \bmod \frac{1}{\epsilon} \neq 0$ is imputing the remaining values with multiple hot deck imputation (proposed by Little and Rubin in 1986) (16), since it preserves the original distribution (proven by Reilly in 1991) (17). If $n \bmod \frac{1}{\epsilon} = \varrho$, the algorithm should run $\binom{n}{\varrho}$ times. An approximation solution is randomly imputing the remaining values several times and then computing the mean of all estimations. Having good performance even for distributions with infinite second moments, the advantages of MoM have received increasing attention over the past decade (15, 18–26). Most of these results should be similar for SM_ϵ . Using a stochastic approach, the variance of MoM is very high, SM_ϵ is obviously almost always a better choice (the only apparent drawback is its computational complexity is $O(n \log n)$, not $O(n)$). In fact, the stratified mean is a type of stratum mean which is related to the stratified sampling. The most similar version was proposed by Takahasi and Wakimoto in 1968 (27), which is stratifying order statistics into several non-overlapping blocks and then computing the mean of one block. The median of means and stratum mean are consistent mean estimators if their asymptotic breakdown points are zero. However, if $\epsilon = \frac{1}{9}$, the biases of the $\text{SM}_{\frac{1}{9}}$ are nearly identical to those of the $\text{WM}_{\frac{1}{9}}$ (Figure 1). More importantly,

Significance Statement

In 1956, Stein initiated the problem of estimating parameters in the presence of an infinite dimensional nuisance shape parameter. A notable example discussed in his seminal work was the estimation of the center of symmetry for an unknown symmetric distribution. However, there is another old and commonly encountered class of distributions that is seldom mentioned in semi-parametric literature: unimodal distributions. These distributions are particularly unique within the semi-parametric framework as their definition is based on inequalities. Here, by classifying distributions through inequalities, a series of sophisticated robust mean estimators are deduced. Nearly all common non-parametric robust location estimators are special cases thereof.

T.L. designed research, performed research, analyzed data, and wrote the paper.

The author declares no competing interest.

¹ To whom correspondence should be addressed. E-mail: tl@biomathematics.org

the bounds confirm that the worst-case performances of WM_ϵ are better than those of TM_ϵ in terms of bias, but the mechanisms of the bias reduction are unclear. This paper aims to define a series of semi-parametric models through inequalities, demonstrate their elegant interrelations and connections to parametric models, and deduce a series of sophisticated robust mean estimators.

Classifying distributions through inequalities

Let $\mathcal{P}_k$ denote the set of all distributions over $\mathbb{R}$ whose moments, from the first to the k th, are all finite. Without loss of generality, all classes discussed in the following are subclasses of the non-parametric class of distributions such that $\mathcal{P}_1^k := \{P \text{ is continuous} \wedge \text{all } P \in \mathcal{P}_k\}$. Besides fully and smoothly parameterized by a Euclidean parameter, or just assuming regularity conditions, there are many ways to classify distributions. After Stein's landmark paper (28) (1956), it soon became clear that the symmetric distribution is a semi-parametric class of distributions, or model (29–31). In 1993, Bickel, Klaassen, Ritov, and Wellner published an influential semiparametric book (32). They systematically classified nearly all common models into three classes: parametric, non-parametric, and semi-parametric. However, there is another old and common class of distributions that receives little attention in semi-parametric literature: the unimodal distribution. It is a very unique semi-parametric model because its definition is based on inequalities, i.e., assuming P is continuous, $(f'(x) > 0 \text{ for } x \leq M) \wedge (f'(x) < 0 \text{ for } x \geq M)$. Let $\mathcal{P}_U$ denote the set of all unimodal distributions. Five parametric distributions in $\mathcal{P}_U$ are interested here: Weibull, gamma, Pareto, lognormal and generalized Gaussian. There was a widespread misbelief that the median is always located between the mean and the mode for an arbitrary unimodal distribution until Runnenburg (1978) and van Zwet (1979) (33, 34) endeavored to determine sufficient conditions under which the inequality holds, thus implying the possibility of its violation (counterexamples see Dharmadhikari and Joag-Dev (1988), Basu and DasGupta (1997), and Abadir (2005)'s papers) (35–37). That means the class of distributions that follow the mean-median-mode inequality is a subclass of $\mathcal{P}_U$. Analogously, a right-skewed distribution follows the quantile inequality if and only if

$$\forall \epsilon_1 \leq \epsilon_2 \leq \frac{1}{2}, \text{SQA}_{\epsilon_1} \geq \text{SQA}_{\epsilon_2}.$$

The left-skewed case is just reversing the inequality; for simplicity, it will be completely omitted in the following discussion, i.e., the right-skewness will always be assumed to be present. Let $\mathcal{P}_Q$ denote the set of all distributions following the quantile inequality. Nearly all common unimodal distributions, including Weibull, gamma, Pareto, lognormal and generalized Gaussian, are in $\mathcal{P}_U \cap \mathcal{P}_Q$ (proven in the following discussion and SI Text). The only minor exceptions occur when the Weibull and gamma distribution are near-symmetric (shown in the SI Text). Unlike the mean-median-mode inequality, whose sufficient conditions are very cumbersome, a necessary and sufficient condition of the quantile inequality is the monotonic property of the bias function of SQA_ϵ with respect to ϵ (proven in the SI Text). The following necessary and sufficient condition hints at the relation between the mean-median-mode inequality and the quantile inequality.

Theorem .1. Let P_Γ^k denote an arbitrary distribution in the set $\mathcal{P}_\Gamma^k$. $P_\Gamma^k \in \mathcal{P}_Q$ if and only if the pdf satisfies the inequality $f(Q(\epsilon)) \geq f(Q(1-\epsilon))$, where $0 \leq \epsilon \leq \frac{1}{2}$ (also assumed in the following discussions), $Q(\epsilon)$ is the quantile function.

Proof. From the definition of quantile inequality, deducing $\frac{Q(\epsilon-\delta)+Q(1-\epsilon+\delta)}{2} \geq \frac{Q(\epsilon)+Q(1-\epsilon)}{2} \Leftrightarrow Q(\epsilon-\delta) - Q(\epsilon) \geq Q(1-\epsilon) - Q(1-\epsilon+\delta) \Leftrightarrow Q'(1-\epsilon) \geq Q'(\epsilon)$, where δ is an infinitesimal quantity. Since the quantile function is the inverse function of the cumulative distribution function (cdf), $Q'(1-\epsilon) \geq Q'(\epsilon) \Leftrightarrow F'(Q(\epsilon)) \geq F'(Q(1-\epsilon))$, the proof is complete by noticing that the derivative of cdf is pdf. $\square$

The mean-median difference $|\mu - m|$ was proposed to measure skewness by Pearson (1895) (38). Bowley (1926) proposed a robust skewness based on the SQA-median difference $|\text{SQA}_\epsilon - m|$ (39). van Zwet (1964) and Groeneveld and Meeden (1984) (40, 41) generalized these measures of skewness and investigated their properties. Suppose P_Γ^k follows the mean-median-mode inequality. Then, the probability density $f(Q(\epsilon))$ on the left side of the median, on average, is greater than the corresponding $f(Q(1-\epsilon))$, since $m < \frac{Q(0)+Q(1)}{2} \Leftrightarrow m - Q(0) < Q(1) - m$. If $Q(\epsilon) > M$, the inequality $f(Q(\epsilon)) \geq f(Q(1-\epsilon))$ holds. The principle can be extended to unimodal-like distributions. Suppose there is a right-skewed continuous multimodal distribution following the mean-median-first mode inequality with many small modes on the right side, the first mode, M , has the greatest probability density and the median is within the first dominant mode, i.e., if $x > m$, $f(m) \geq f(x)$. Then, if $Q(\epsilon) > M$, the inequality $f(Q(\epsilon)) \geq f(Q(1-\epsilon))$ will also holds.

Furthermore, most common right-skewed distributions are partial-bounded. This implies the convex decreasing behavior of the SQA function when $\epsilon \rightarrow 0$. If assuming convexity further, the second quantile inequality can be defined as the following for a right-skewed distribution plus the quantile inequality,

$$\forall \epsilon_1 \leq \epsilon_2 \leq \epsilon_3 \leq \frac{1}{2}, \frac{\text{SQA}_{\epsilon_1} - \text{SQA}_{\epsilon_2}}{\epsilon_2 - \epsilon_1} \geq \frac{\text{SQA}_{\epsilon_2} - \text{SQA}_{\epsilon_3}}{\epsilon_3 - \epsilon_2}.$$

An equivalent expression is $\frac{d^2 \text{SQA}}{d\epsilon^2} \geq 0 \wedge \frac{d \text{SQA}}{d\epsilon} \leq 0$. Analogously, the ν th quantile inequality can be defined as $(-1)^\nu \frac{d^\nu \text{SQA}}{d\epsilon^\nu} \geq 0 \wedge \dots \wedge -\frac{d \text{SQA}}{d\epsilon} \geq 0$. The second and third quantile inequalities are also generally valid for common unimodal distributions (shown in the SI Text). Let $\mathcal{P}_{Q_\nu}$ denote the set of all distributions following the ν th quantile inequality. The following four theorems can be used to quickly identify parametric distributions in $\mathcal{P}_{Q_\nu}$ without solving the derivative.

Theorem .2. Any symmetric distribution with a finite second moment follows the ν th quantile inequality.

Proof. The assertion follows from the fact that for any symmetric distribution with a finite second moment, all symmetric quantile averages coincide. Therefore, the SQA function is always a horizontal line; the ν th order derivative is zero. $\square$

As a consequence of Theorem .2 and the fact that generalized Gaussian distribution is symmetric around the median, it follows the ν th quantile inequality.

Theorem .3. Any continuous right skewed distribution whose Q satisfies $Q^{(\nu)}(p) \geq 0 \wedge \dots \wedge Q^{(i)}(p) \geq 0 \wedge Q^{(2)}(p) \geq 0$, $i \bmod 2 = 0$, follows the ν th quantile inequality.

104 *Proof.* Let $\text{SQA}(\epsilon) = \frac{1}{2}(Q(\epsilon) + Q(1 - \epsilon))$, then $(-1)^j \frac{d^j \text{SQA}}{d\epsilon^j} =$
 105 $\frac{1}{2}((-1)^j Q^j(\epsilon) + Q^j(1 - \epsilon))$, $\nu \geq j \geq 1$, when $j \bmod 2 = 0$,
 106 $(-1)^j \frac{d^j \text{SQA}}{d\epsilon^j} \geq 0$, when $j \bmod 2 = 1$, $(-1)^j \frac{d^j \text{SQA}}{d\epsilon^j} = \frac{1}{2}(Q^j(1 - \epsilon) - Q^j(\epsilon)) \geq 0$, since $(j + 1) \bmod 2 = 0$, $Q^{(j+1)}(\epsilon) \geq 0$. $\square$

108 It is now trivial to prove that the Pareto distribution follows
 109 the ν th quantile inequality since the quantile function of the
 110 Pareto distribution is $Q(p) = x_m(1 - p)^{-\frac{1}{\alpha}}$, $x_m > 0$, $\alpha > 0$,
 111 $Q^{(\nu)}(p) \geq 0$ according to the chain rule.

112 **Theorem .4.** *A right-skewed continuous distribution with a*
 113 *monotonic decreasing pdf follows the quantile inequality.*

114 *Proof.* A monotonic decreasing pdf means $f'(x) = F^{(2)}(x) \leq$
 115 0 . Since $Q'(p) \geq 0$, let $x = Q(F(x))$, then by differenti-
 116 ating both sides of the equation twice, one can obtain $0 =$
 117 $Q^{(2)}(F(x))(F'(x))^2 + Q'(F(x))F^{(2)}(x) \Leftrightarrow Q^{(2)}(F(x)) =$
 118 $-\frac{Q'(F(x))F^{(2)}(x)}{(F'(x))^2} \geq 0$. The desired result is from Theorem
 119 .3. $\square$

120 Theorem .4 gives an interesting insight into the relation
 121 between modality and quantile inequality. According to the
 122 conventional definition, a distribution with a monotonic pdf is
 123 still a unimodal distribution. However, within the interval sup-
 124 ported, the mode number is zero. In fact, the number of modes
 125 and their magnitudes are closely related to the possibility of
 126 the validity of quantile inequality, even though counterexamples
 127 can always be constructed. A quantile inequality proof
 128 can be done for the gamma distributions when $\alpha \leq 1$ since the
 129 pdf of the gamma distribution is $f(x) = \frac{\lambda^{-\alpha} x^{\alpha-1} e^{-\frac{x}{\lambda}}}{\Gamma(\alpha)}$, $x \geq 0$,
 130 $\lambda > 0$, $\alpha > 0$, which is a product of two monotonic decreasing
 131 functions under constraints. For $\alpha > 1$, the proof is hard
 132 (42–44), numerical results show that the quantile inequality
 133 is valid until $\alpha > 140$, but it is instructive to consider that
 134 when $\alpha \rightarrow \infty$ the gamma distribution converges to a Gaussian
 135 distribution with mean $\mu = \alpha\lambda$ and variance $\sigma = \alpha\lambda^2$.

136 **Theorem .5.** *If transforming a symmetric unimodal random*
 137 *variable X with a function $\chi(x)$ such that $\frac{d^2 \chi}{dx^2} \geq 0 \wedge \frac{d\chi}{dx} \geq 0$*
 138 *over the interval supported, then the monotonic transformed*
 139 *distribution follows the quantile inequality.*

140 *Proof.* Let $\chi\text{SQA}(\epsilon) = \frac{1}{2}(\chi(Q(\epsilon)) + \chi(Q(1 - \epsilon)))$, then,
 141 $\frac{d\chi\text{SQA}}{d\epsilon} = \frac{1}{2}(\chi'(Q(\epsilon))Q'(\epsilon) - \chi'(Q(1 - \epsilon))Q'(1 - \epsilon)) =$
 142 $\frac{1}{2}Q'(\epsilon)(\chi'(Q(\epsilon)) - \chi'(Q(1 - \epsilon))) \leq 0$, since for a symmet-
 143 ric distribution, $m - Q(\epsilon) = Q(1 - \epsilon) - m$, differentiat-
 144 ing both sides, $-Q'(\epsilon) = -Q'(1 - \epsilon)$, $Q'(\epsilon) \geq 0$, $\chi^{(2)} \geq$
 145 0 . Notably, differentiating twice, $Q^{(2)}(\epsilon) = -Q^{(2)}(1 - \epsilon)$,
 146 $\frac{d^{(2)}\chi\text{SQA}}{d\epsilon^{(2)}} = \frac{1}{2}((\chi^{(2)}(Q(\epsilon)) + \chi^{(2)}(Q(1 - \epsilon)))(Q'(\epsilon))^2 +$
 147 $\frac{1}{2}((\chi'(Q(\epsilon)) - \chi'(Q(1 - \epsilon)))Q^{(2)}(\epsilon))$. The sign of $\frac{d^{(2)}\chi\text{SQA}}{d\epsilon^{(2)}}$
 148 depends on $Q^{(2)}(\epsilon)$. If assuming $Q^{(2)}(\epsilon) \leq 0$, the distribution
 149 follows the second quantile inequality. $\square$

150 The mean-median-mode inequality for distributions of the
 151 powers and roots of the variates of a given distribution was
 152 investigated by Henry Rietz in 1927 (45), but the most trivial
 153 solution is the exponential transformation since the derivatives
 154 are always positive. An application of Theorem .5 is that the
 155 lognormal distribution also follows the quantile inequality as
 156 it is exponentially transformed from the Gaussian distribution

whose $Q^{(2)}(\epsilon) = -2\sqrt{2\pi}\sigma e^{2\text{erfc}^{-1}(2\epsilon)^2} \text{erfc}^{-1}(2\epsilon) \leq 0$ (so, the
 lognormal distribution follows the second quantile inequality).

Remarkably, the bias bound of the symmetric quantile
 average for $\mathcal{P}_U \cap \mathcal{P}_2$ derived by Bernard et al. (2020)
 (2) is also a monotonic function (proven in the SI Text).
 This result implies, even if the quantile inequality is vi-
 olated, e.g., for a right-skewed unimodal distribution, if
 $\exists \epsilon_1 \leq \epsilon_2 \leq \epsilon_3 \leq \frac{1}{2}$, $\text{SQA}_{\epsilon_2} \geq \text{SQA}_{\epsilon_3} \geq \text{SQA}_{\epsilon_1}$, SQA_{ϵ_2} will not
 be too far away from SQA_{ϵ_1} , since $\sup_{P \in \mathcal{P}_U \cap \mathcal{P}_2} (\text{SQA}_{\epsilon_1}) >$
 $\sup_{P \in \mathcal{P}_U \cap \mathcal{P}_2} (\text{SQA}_{\epsilon_2}) > \sup_{P \in \mathcal{P}_U \cap \mathcal{P}_2} (\text{SQA}_{\epsilon_3})$.

Symmetric weighted averages and their related inequalities

ϵ -symmetric trimmed mean, ϵ -symmetric Winsorized mean,
 and ϵ -stratified mean are all weighted averages according to
 Daniell's definition (5). More specifically, they are all symmet-
 ric weighted averages, which is defined as

$$\text{SWA}_{\epsilon,n} := \frac{\sum_{i=1}^{\frac{n}{2}} w_i \frac{X_i + X_{n-i+1}}{2}}{\sum_{i=1}^{\frac{n}{2}} w_i},$$

where w_i s are the weights applied to symmetric quantile aver-
 ages according to the definition of the corresponding weighted
 average. For example, for the ϵ -symmetric trimmed mean,
 $w_i = \begin{cases} 0, & i < n\epsilon \\ 1, & i \geq n\epsilon \end{cases}$. Mean ($\lim_{\epsilon \rightarrow 0} \text{TM}_{\epsilon}$) and median ($\text{TM}_{\frac{1}{2}}$)
 are just two special cases of trimmed mean. The bias bound of
 the trimmed mean is also monotonic for $\mathcal{P}_U \cap \mathcal{P}_2$ (proven in the
 SI Text using the formula provided in Bernard et al.'s paper) (2).
 So far, it appears clear that the bias of an estimator is closely
 related to its degree of robustness. Analogous to the quantile
 inequality, for a right-skewed unimodal distribution, often,
 the mean, ϵ -symmetric trimmed mean, and median occur in
 reverse alphabetical order, $\forall \epsilon_1 \leq \epsilon_2 \leq \frac{1}{2}$, $\text{TM}_{\epsilon_1} \geq \text{TM}_{\epsilon_2}$. A
 necessary and sufficient condition of the trimming inequality is
 also the monotonic behavior of the bias of the trimmed mean
 as a function of the breakdown point ϵ (proven in the SI Text).
 Quantile inequality is a sufficient condition (proven in the SI
 Text), but it is not necessary.

Theorem .6. *For a right-skewed continuous distribution fol-*
lowing the trimming inequality, the symmetric quantile average
is always greater or equal to the corresponding trimmed mean
with the same breakdown point ϵ .

Proof. By deducing

$$\begin{aligned} \frac{1}{1-2\epsilon+2\delta} \int_{\epsilon-\delta}^{1-\epsilon+\delta} Q(u) du &\geq \frac{1}{1-2\epsilon} \int_{\epsilon}^{1-\epsilon} Q(u) du \Leftrightarrow \\ \int_{\epsilon-\delta}^{1-\epsilon+\delta} Q(u) du - \frac{1-2\epsilon+2\delta}{1-2\epsilon} \int_{\epsilon}^{1-\epsilon} Q(u) du &\geq 0 \Leftrightarrow \\ \int_{1-\epsilon}^{1-\epsilon+\delta} Q(u) du + \int_{\epsilon-\delta}^{\epsilon} Q(u) du - \frac{2\delta}{1-2\epsilon} \int_{\epsilon}^{1-\epsilon} Q(u) du &\geq 0 \Leftrightarrow \\ \frac{1}{2\delta} \left(\int_{1-\epsilon}^{1-\epsilon+\delta} Q(u) du + \int_{\epsilon-\delta}^{\epsilon} Q(u) du \right) &\geq \frac{1}{1-2\epsilon} \int_{\epsilon}^{1-\epsilon} Q(u) du \end{aligned}$$

and noting that $\lim_{\delta \rightarrow 0} \frac{1}{2\delta} \left(\int_{1-\epsilon}^{1-\epsilon+\delta} Q(u) du + \int_{\epsilon-\delta}^{\epsilon} Q(u) du \right) = \frac{Q(\epsilon) + Q(1-\epsilon)}{2}$, the proof is complete. $\square$

Additionally, the quantile inequality is a sufficient condition
 of the Winsorization inequality (proven in the SI Text). A
 similar result can be obtained in the following theorem.

Theorem .7. *For a right-skewed continuous distribution fol-*
lowing the trimming inequality, the Winsorized mean is always

greater or equal to the corresponding trimmed mean with the same breakdown point ϵ .

Proof. Continue the above deduction,

$$\frac{Q(\epsilon) + Q(1-\epsilon)}{2} \geq \frac{1}{1-2\epsilon} \int_{\epsilon}^{1-\epsilon} Q(u) du \Leftrightarrow$$

$$\epsilon(Q(\epsilon) + Q(1-\epsilon)) \geq \left(\frac{2\epsilon}{1-2\epsilon}\right) \int_{\epsilon}^{1-\epsilon} Q(u) du \Leftrightarrow$$

$$\left(1 - \frac{1}{1-2\epsilon}\right) \int_{\epsilon}^{1-\epsilon} Q(u) du + \epsilon(Q(\epsilon) + Q(1-\epsilon)) \geq 0 \Leftrightarrow$$

$$\int_{\epsilon}^{1-\epsilon} Q(u) du + \epsilon(Q(\epsilon) + Q(1-\epsilon)) \geq \frac{1}{1-2\epsilon} \int_{\epsilon}^{1-\epsilon} Q(u) du,$$
the proof is complete. $\square$

Theorem .7 unequivocally establishes that, for a distribution that follows the trimming inequality (also quantile inequality), the Winsorized mean is always less biased than the trimmed mean. If one wants to construct a robust SWA following the mean-symmetric weighted average-median inequality. According to the definition of the quantile inequality, for a right-skewed or symmetric distribution, $SWA \geq m$. So, the condition that SWA is between m and μ is $\mu \geq SWA$. Let $B_i = \int_{i\epsilon}^{(i+1)\epsilon} SQA(u) du$, $ka = k\epsilon + b$, from the quantile inequality, $-\frac{dSQA}{da} \geq 0 \Rightarrow \forall a \leq 2a \leq \frac{1}{2}$, $-\frac{(SQA(2a) - SQA(a))}{a} \geq 0 \Rightarrow B_i - B_{i+1} \geq 0$, from the second quantile inequality $\frac{d^2SQA}{da^2} \geq 0 \Rightarrow \forall a \leq 2a \leq 3a \leq \frac{1}{2}$, $\frac{1}{a} \left(\frac{SQA(3a) - SQA(2a)}{a} - \frac{SQA(2a) - SQA(a)}{a} \right) \geq 0 \Rightarrow B_i - 2B_{i+1} + B_{i+2} \geq 0$. Let $B_i = B_0$, then, from the quantile inequality, another version of ϵ -Winsorized mean is defined here for comparison in the SI Dataset S1 as

$$WM2_{\epsilon,n} := \frac{1}{n} \left(\sum_{i=n\epsilon+1}^{(1-\epsilon)n} X_i + \sum_{i=n\epsilon+1}^{2n\epsilon} X_i + \sum_{i=(1-2\epsilon)n+1}^{(1-\epsilon)n} X_i \right),$$

which is double weighting the leftest and rightest blocks having the size ϵn . SM_{ϵ} can be seen as replacing the two blocks, $B_i + B_{i+2}$ with one block $2B_{i+1}$. From the ν th inequality, the recurrence relation of the derivatives naturally produces the alternating binomial coefficients,

$$(-1)^{\nu} \frac{d^{\nu} SQA}{da^{\nu}} \geq 0 \Rightarrow \forall a \leq \dots \leq (\nu+1)a \leq \frac{1}{2},$$

$$(-1)^{\nu} \frac{1}{a} \left(\frac{\frac{SQA(\nu a + a)}{a} - \dots - \frac{SQA(2a)}{a}}{a} - \frac{\frac{SQA(\nu a)}{a} - \dots - \frac{SQA(a)}{a}}{a} \right)$$

$$\geq 0 \Leftrightarrow (-1)^{\nu} \frac{1}{a^{\nu}} \left(\sum_{j=0}^{\nu} (-1)^j \binom{\nu}{j} SQA((\nu-j+1)a) \right) \geq 0$$

$$\Rightarrow \sum_{j=0}^{\nu} (-1)^j \binom{\nu}{j} B_{i+j} \geq 0.$$

Based on the ν th quantile inequality, the ϵ -binomial mean is introduced as

$$BM_{\nu,\epsilon,n} := \frac{1}{n} \left(\sum_{i=1}^{\frac{1}{2}\epsilon^{-1}(\nu+1)-1} \sum_{j=0}^{\nu} \left(1 - (-1)^j \binom{\nu}{j} \right) \mathfrak{B}_{ij} \right),$$

where $\mathfrak{B}_{ij} = \sum_{l=n\epsilon(j+(i-1)(\nu+1)+1)}^{n\epsilon(j+(i-1)(\nu+1)+1)} (X_l + X_{n-l+1})$, $\frac{1}{2}\epsilon^{-1}(\nu+1) \in \mathbb{N}$. If ν is not indicated, it is default as $\frac{1}{2}\epsilon^{-1} - 1$. The solution for $n \bmod \epsilon^{-1} \neq 0$ is the same as that in the stratified mean. $BM_{\nu=2,\epsilon}$ is identical to SM_{ϵ} , if the two ϵ s are the same. The reason that $SM_{\frac{1}{9}}$ has similar biases as

the corresponding $WM_{\frac{1}{9}}$ is the Winsorized mean is using two single quantiles to replace the trimmed parts, not two blocks. If using single quantiles, based on the second quantile inequality, the stratified quantile mean can be defined as

$$SQM_{\epsilon,n} := 4\epsilon \sum_{i=1}^{\frac{1}{4\epsilon}} \frac{1}{2} (\hat{Q}_n((2i-1)\epsilon) + \hat{Q}_n(1 - (2i-1)\epsilon)),$$

where $\frac{1}{\epsilon} \bmod 4 = 0$, $\hat{Q}_n(p)$ is the empirical quantile function. $SQM_{\frac{1}{4}}$ is the Tukey's midhinge (46). The biases of $SQM_{\frac{1}{8}}$ are also very close to those of $BM_{\frac{1}{8}}$ (Figure 1), which is based on the third quantile inequality.

Hodges-Lehmann inequality and U -quantile inequality

There are two methods to compute the H-L estimator, median of pairwise means and Wilcoxon score, whose time complexities are $O(n^2)$ and $O(n \log(n))$, respectively. The H-L estimator is a very unique robust location estimator because it is not a symmetric weighted average. Instead, it is a symmetric weighted H-L mean, which is defined as

$$SWHLM_{k,\epsilon,n} := SWA_{\epsilon,n} \left((hl_k(X_{N_1}, \dots, X_{N_k}))_{N=1}^{\binom{n}{k}} \right),$$

where $hl_k = \frac{1}{k} \sum_{i=1}^k x_i$, $SWA_{\epsilon,n}(Y)$ denotes the ϵ -symmetric weighted average with the sequence $(hl_k(X_{N_1}, \dots, X_{N_k}))_{N=1}^{\binom{n}{k}}$ as an input. Set the SWA as TM, the $k=2$ case is a version of the Wilcoxon one-sample statistic investigated by Saleh in 1976 (47), which is named as trimmed H-L mean (Figure 1). The hl_k kernel was pointed by Janssen, Serfling and Veraverbeke in 1987 (48) and also Shulenin in 2020 (49). The hl_2 kernel distribution has a probability density function $f_{hl_2}(x) = \int_0^{2x} 2f(t)f(2x-t)dt$ (a result after the transformation of variables), the support of the original distribution is assumed to be $[0, \infty)$ for simplicity. The expected value is the solution of $\int_0^{H-L} (f_{hl_2}(s))ds = \frac{1}{2}$. Due to the complexity of the equation, analytically proving the validity of the mean-H-L-median inequality for a distribution is hard. As an example, for the exponential distribution, $f_{hl_2}(x) = 4\lambda^{-2}xe^{-2\lambda^{-1}x}$, $E[H-L] = \frac{-W_{-1}(-\frac{1}{2\epsilon})-1}{2} \lambda \approx 0.839\lambda$, where W is the Lambert W function. Analogous to the trimming inequality, the Hodges-Lehmann inequality can be defined as $\forall n \geq k_2 \geq k_1 \geq 1, mHLM_{k_2} \geq mHLM_{k_1}$, where $mHLM_k$ is setting the SWA as median. Since $mHLM_{k=1} = m$, $mHLM_{k=2} = H-L$, $mHLM_{k=\infty} = \mu$, if a distribution follows the H-L inequality, it also follows the mean-H-L-median inequality. Furthermore, the Hodges-Lehmann inequality is a special case of the U -quantile inequality, i.e.,

$$(\forall n \geq k_2 \geq k_1 \geq 1, SQHLM_{k_2,\epsilon} \geq SQHLM_{k_1,\epsilon}) \wedge$$

$$(\forall n \geq k_2 \geq k_1 \geq 1, SQHLM_{k_2,\epsilon} \leq SQHLM_{k_1,\epsilon}).$$

The direction of the inequality depends on SQA_{ϵ} and μ , since $SQHLM_{k=1,\epsilon} = SQA_{\epsilon}$.

Theorem .8. U -quantile inequality implies quantile inequality.

Proof. Suppose $n \bmod 2 = 0$, $n \rightarrow \infty$, the quantile inequality implies that $\frac{1}{2}(X_1 + X_n) \geq \dots \geq \frac{1}{2}(X_i + X_{n-i+1}) \geq \dots \geq \frac{1}{2}(X_{\frac{n}{2}} + X_{\frac{n}{2}+1})$. Let $\tilde{\epsilon} = \frac{i}{n}$, when $\tilde{\epsilon} \rightarrow 0$, $SQHLM_{k_2,\tilde{\epsilon}} \leq SQHLM_{k_1,\tilde{\epsilon}}$ is equivalent to the quantile inequality, since

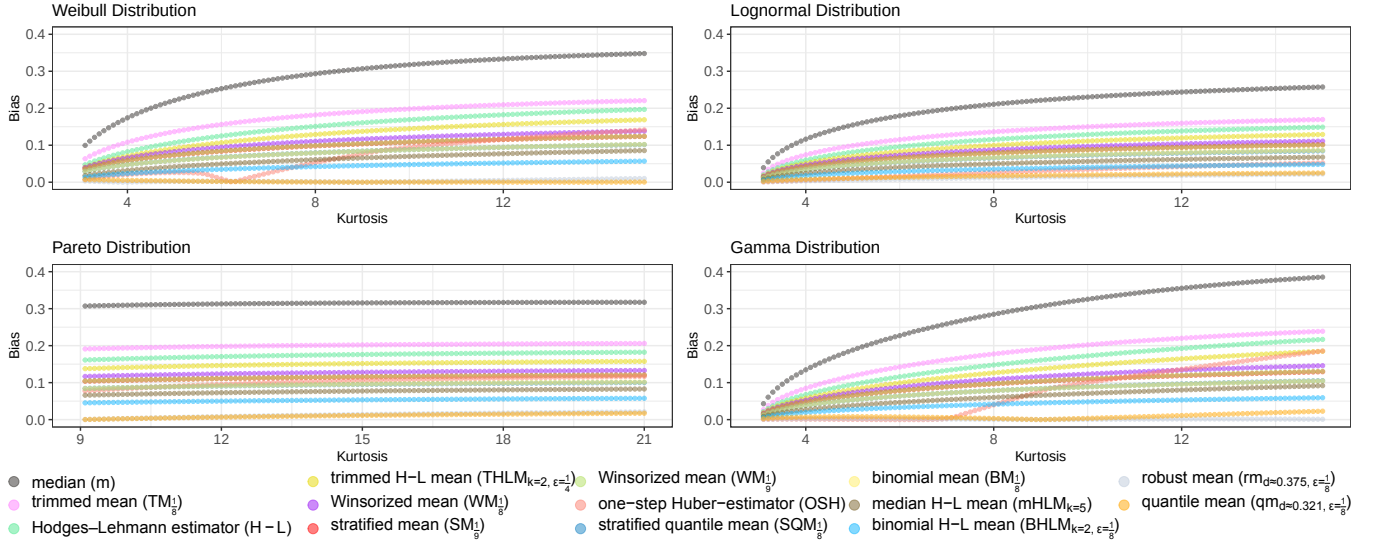


Fig. 1. Standardized asymptotic biases (with respect to μ) of fourteen robust location estimators (including three that detailed in another relevant paper for better comparison) in four two-parameter skewed unimodal distributions as a function of the kurtosis.

SQHLM $_{k=j, \epsilon \rightarrow 0, n} = \frac{1}{2j} \left(\sum_{i=1}^j (X_i + X_{n-i+1}) \right)$ and Theorem 6 implies that $\mu \leq \text{SQA}_{\epsilon}$. $\square$

Be aware that the U -quantile inequality itself does not assume any inequality within the hl_k kernel distribution. The $hl_{k=n-1}$ kernel distribution has n elements, and the order is the same as the original distribution, so it follows the quantile inequality if and only if the original distribution follows the quantile inequality. If assuming symmetry, the result is trivial since the k -fold convolutions of a symmetric distribution is also symmetric (50). Proving other cases is challenging. For example, $f'_{hl_2}(x) = 4f(2x)f(0) + \int_0^{2x} 4f(t)f'(2x-t)dt$, the strict negative of $f'_{hl_2}(x)$ is not guaranteed if just assuming $f'(x) < 0$, so, even the original distribution is monotonic, the hl_2 kernel distribution might be non-monotonic. Also, unlike the pairwise difference distribution, if the original distribution is unimodal, the pairwise mean distribution might be non-unimodal, as a counterexample was given by Chung in 1953 and mentioned by Hodges and Lehmann (51, 52). If all hl_k kernel distributions, $n \geq k \geq 2$, follow the ν th quantile inequality, the distribution follows the ν th U -quantile inequality. From that, binomial H-L mean (set the SWA as BM) might be the bias-optimum choice without specific parametric assumptions (Figure 1), although its maximum breakdown point is ≈ 0.065 . Having similar breakdown points as $\text{BM}_{\frac{1}{8}}$, the biases of $\text{mHLM}_{k=5}$ are smaller (Figure 1), but the computational cost is much higher.

Methods

This section describes the methods of generating Figure 1.

The hardness of computing the bias bounds of complex estimators under strict assumptions and the discrepancies between the average and worst-case performances of some estimators motivate the definition of average asymptotic bias.

Definition .1 (Average asymptotic bias). For a single-parameter distribution, the average asymptotic bias (AAB) is just the standardized asymptotic bias $\frac{|\hat{\theta} - \theta|}{\sigma}$, where $\hat{\theta}$ is the estimation of θ and σ is the standard deviation of the distribution. For a two-parameter

distribution, the first step is setting the lower bound of the kurtosis range of interest κ_l . Then, the average asymptotic bias is defined as

$$\text{AAB}_{\hat{\theta}} := \frac{1}{C} \sum_{\substack{\delta + \kappa_l \leq \kappa \leq C\delta + \kappa_l \\ \kappa \text{ is a multiple of } \delta}} E_{x|\kappa} \left[\frac{|\hat{\theta} - \theta|}{\sigma} \right]$$

where κ is the kurtosis specified, $E_{x|\kappa}$ denotes the expected value over the distribution $P(x|\kappa)$.

Here, $\delta = 0.1$, $C = 120$, $\kappa_l = \lceil \kappa_{min} \rceil$, κ_{min} is the minimum kurtosis defined in the distribution. For the Weibull, gamma, and lognormal distributions, the kurtosis range is from 3 to 15 (there are two shape parameter solutions for the Weibull distribution, the lower one is used here). For the Pareto distribution, the range is from 9 to 21.

Standardization is crucial for comparison among different distributions. Currently, there are many choices, e.g., Gauss used the root mean square deviation from the mode, whereas mean absolute deviation and standard deviation are two common choices. Standard deviation is used here because of its central role in standard error estimation.

The asymptotic results were approximated by a quasi-Monte Carlo study (53, 54) based on generating a large quasi-random sample with sample size 1.8 million for corresponding distributions and quasi-subsampling the sample 1.8 million times (55) to approximate the distributions of the kernels of the symmetric weighted average Hodges-Lehmann mean. The accuracies were checked by comparing the results to the asymptotic values for the exponential distribution (errors), showing that the errors were all smaller than 0.001σ . The simulation results and related codes are all included in the SI Dataset S1 and [GitHub](#).

Data Availability. Data for Figure 1 are given in SI Dataset S1. All codes have been deposited in [GitHub](#).

ACKNOWLEDGMENTS. I gratefully acknowledge the valuable comments by the editor which substantially improved the clarity and quality of this paper.

1. CF Gauss, *Theoria combinationis observationum erroribus minimis obnoxiae*. (Henricus Dieterich), (1823).
2. C Bernard, R Kazzi, S Vanduffel, Range value-at-risk bounds for unimodal distributions under partial information. *Insur. Math. Econ.* **94**, 9–24 (2020).
3. S Karlin, A Novikoff, Generalized convex inequalities. *Pac. J. Math.* **13**, 1251–1279 (1963).

- 288 4. L Li, H Shao, R Wang, J Yang, Worst-case range value-at-risk with partial information. *SIAM J.*
289 *on Financial Math.* **9**, 190–218 (2018).
- 290 5. P Daniell, Observations weighted according to order. *Am. J. Math.* **42**, 222–236 (1920).
- 291 6. JW Tukey, A survey of sampling from contaminated distributions in *Contributions to probability*
292 *and statistics*. (Stanford University Press), pp. 448–485 (1960).
- 293 7. WJ Dixon, Simplified Estimation from Censored Normal Samples. *The Annals Math. Stat.* **31**,
294 385 – 391 (1960).
- 295 8. M Bieniek, Comparison of the bias of trimmed and winsorized means. *Commun. Stat. Methods*
296 **45**, 6641–6650 (2016).
- 297 9. K Danielak, T Rychlik, Theory & methods: Exact bounds for the bias of trimmed means. *Aust.*
298 *& New Zealand J. Stat.* **45**, 83–96 (2003).
- 299 10. J Hodges Jr, E Lehmann, Estimates of location based on rank tests. *The Annals Math. Stat.*
300 **34**, 598–611 (1963).
- 301 11. F Wilcoxon, Individual comparisons by ranking methods. *Biom. Bull.* **1**, 80–83 (1945).
- 302 12. AS Nemirovskij, DB Yudin, *Problem complexity and method efficiency in optimization*. (Wiley-
303 Interscience), (1983).
- 304 13. MR Jerrum, LG Valiant, VV Vazirani, Random generation of combinatorial structures from a
305 uniform distribution. *Theor. computer science* **43**, 169–188 (1986).
- 306 14. N Alon, Y Matias, M Szegedy, The space complexity of approximating the frequency moments
307 in *Proceedings of the twenty-eighth annual ACM symposium on Theory of computing*. pp.
308 20–29 (1996).
- 309 15. L Devroye, M Lerasle, G Lugosi, RI Oliveira, Sub-gaussian mean estimators. *The Annals Stat.*
310 **44**, 2695–2725 (2016).
- 311 16. RJ Little, DB Rubin, *Statistical analysis with missing data*. (John Wiley & Sons) Vol. 793,
312 (2019).
- 313 17. M Reilly, *Semi-parametric methods of dealing with missing or surrogate covariate data*. (Uni-
314 versity of Washington), (1991).
- 315 18. PL Bühlmann, Bagging, subagging and bragging for improving some prediction algorithms
316 in *Research report/Seminar für Statistik, Eidgenössische Technische Hochschule (ETH)*.
317 (Seminar für Statistik, Eidgenössische Technische Hochschule (ETH), Zürich), Vol. 113,
318 (2003).
- 319 19. JY Audibert, O Catoni, Robust linear least squares regression. *The Annals Stat.* **39**, 2766–2794
320 (2011).
- 321 20. O Catoni, Challenging the empirical mean and empirical variance: a deviation study. *Annales*
322 *de l'IHP Probab. et statistiques* **48**, 1148–1185 (2012).
- 323 21. D Hsu, S Sabato, Heavy-tailed regression with a generalized median-of-means in *International*
324 *Conference on Machine Learning*. (PMLR), pp. 37–45 (2014).
- 325 22. S Minsker, Geometric median and robust estimation in banach spaces. *Bernoulli* **21**, 2308–
326 2335 (2015).
- 327 23. C Brownlees, E Joly, G Lugosi, Empirical risk minimization for heavy-tailed losses. *The Annals*
328 *Stat.* **43**, 2507–2536 (2015).
- 329 24. D Hsu, S Sabato, Loss minimization and parameter estimation with heavy tails. *The J. Mach.*
330 *Learn. Res.* **17**, 543–582 (2016).
- 331 25. E Joly, G Lugosi, Robust estimation of u-statistics. *Stoch. Process. their Appl.* **126**, 3760–3773
332 (2016).
- 333 26. P Laforge, S Cléménçon, P Bertail, On medians of (randomized) pairwise means in *Interna-*
334 *tional Conference on Machine Learning*. (PMLR), pp. 1272–1281 (2019).
- 335 27. K Takahasi, K Wakimoto, On unbiased estimates of the population mean based on the sample
336 stratified by means of ordering. *Annals institute statistical mathematics* **20**, 1–31 (1968).
- 337 28. C Stein, , et al., Efficient nonparametric testing and estimation in *Proceedings of the third*
338 *Berkeley symposium on mathematical statistics and probability*. Vol. 1, pp. 187–195 (1956).
- 339 29. C Van Eeden, Efficiency-robust estimation of location. *The Annals Math. Stat.* **41**, 172–181
340 (1970).
- 341 30. K Takeuchi, A uniformly asymptotically efficient estimator of a location parameter. *J. Am. Stat.*
342 *Assoc.* **66**, 292–301 (1971).
- 343 31. PJ Bickel, On adaptive estimation. *The Annals Stat.* **10**, 647–671 (1982).
- 344 32. P Bickel, CA Klaassen, Y Ritov, JA Wellner, *Efficient and adaptive estimation for semiparamet-*
345 *ric models*. (Springer) Vol. 4, (1993).
- 346 33. JT Runnenburg, Mean, median, mode. *Stat. Neerlandica* **32**, 73–79 (1978).
- 347 34. Wv Zwet, Mean, median, mode ii. *Stat. Neerlandica* **33**, 1–5 (1979).
- 348 35. S Basu, A DasGupta, The mean, median, and mode of unimodal distributions: a characteriza-
349 tion. *Theory Probab. & Its Appl.* **41**, 210–223 (1997).
- 350 36. S Dharmadhikari, K Joag-Dev, *Unimodality, convexity, and applications*. (Elsevier), (1988).
- 351 37. KM Abadir, The mean-median-mode inequality: Counterexamples. *Econom. Theory* **21**,
352 477–482 (2005).
- 353 38. K Pearson, X. contributions to the mathematical theory of evolution.—ii. skew variation in
354 homogeneous material. *Philos. Transactions Royal Soc. London.(A.)* **186**, 343–414 (1895).
- 355 39. AL Bowley, *Elements of statistics*. (King) No. 8, (1926).
- 356 40. W van Zwet, Convex transformations: A new approach to skewness and kurtosis in *Selected*
357 *Works of Willem van Zwet*. (Springer), pp. 3–11 (2012).
- 358 41. RA Groeneveld, G Meeden, Measuring skewness and kurtosis. *J. Royal Stat. Soc. Ser. D*
359 *(The Stat.* **33**, 391–399 (1984).
- 360 42. MA Chaudhry, SM Zubair, *On a class of incomplete gamma functions with applications*.
361 (Chapman and Hall/CRC), (2001).
- 362 43. E Furman, R Zitikis, A monotonicity property of the composition of regularized and inverted-
363 regularized gamma functions with applications. *J. Math. Analysis Appl.* **348**, 971–976 (2008).
- 364 44. W Chojnacki, Some monotonicity and limit results for the regularised incomplete gamma
365 function. *Annales Polonici Math.* **94**, 283–291 (2008).
- 366 45. H Rietz, On certain properties of frequency distributions of the powers and roots of the variates
367 of a given distribution. *Proc. Natl. Acad. Sci.* **13**, 817–820 (1927).
- 368 46. JW Tukey, , et al., *Exploratory data analysis*. (Reading, MA) Vol. 2, (1977).
- 369 47. A Ehsanes Saleh, Hodges-lehmann estimate of the location parameter in censored samples.
370 *Annals Inst. Stat. Math.* **28**, 235–247 (1976).
- 371 48. P Janssen, R Serfling, N Veraverbeke, Asymptotic normality of u-statistics based on trimmed
samples. *J. statistical planning inference* **16**, 63–74 (1987).
49. VP Shulenin, Robustness properties of the modified hodges– lehmann estimates. *Russ. Phys.*
373 *J.* **63**, 574–590 (2020).
50. RG Laha, On a class of unimodal distributions. *Proc. Am. Math. Soc.* **12**, 181–184 (1961).
51. J Hodges, E Lehmann, Matching in paired comparisons. *The Annals Math. Stat.* **25**, 787–791
376 (1954).
52. K Chung, Sur les lois de probabilité unimodales. *COMPTEs RENDUS HEBDOMADAIRES*
378 *DES SEANCES DE L ACADEMIE DES SCIENCES* **236**, 583–584 (1953).
53. RD Richtmyer, A non-random sampling method, based on congruences, for" monte carlo"
380 problems, (New York Univ., New York. Atomic Energy Commission Computing and Applied . . .),
381 Technical report (1958).
54. IM Sobol', On the distribution of points in a cube and the approximate evaluation of integrals.
383 *Zhurnal Vychislitel'noi Matematiki i Matematicheskoi Fiziki* **7**, 784–802 (1967).
55. KA Do, P Hall, Quasi-random resampling for the bootstrap. *Stat. Comput.* **1**, 13–22 (1991).